

Matthew Marsicano

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EDUCATION

Northeastern University

Expected May 2027

Bachelor of Science in Electrical Engineering, **GPA: 3.95**

Boston, MA

Coursework: Electronic Design, Linear Systems, Electromagnetics, Networked XR, Digital Design

Organizations: Generate Product Development, Forge (Silicon Design Director), IEEE

SKILLS

Analog IC: Cadence Virtuoso (Spectre, ADE), Xschem, Ngspice, KLayout, SKY130 PDK, DRC/LVS

Lab & Test: VNA, spectrum analyzer, signal generator, noise figure, phase noise, EVM, P1dB/IP3

RF & PCB: Altium, Allegro, AWR | **Programming:** SystemVerilog, SCPI, Python, MATLAB,

PROFESSIONAL EXPERIENCE

Google

May 2026 - Aug 2026

Hardware Engineer Intern

Chicago, IL

- Characterized PCB Dk/Df and copper up to 20 GHz by DCR, TDR, and S-parameters measurements.
- Designed coupon structures to IEEE-370 with Delta L+3, Beatty resonators, and 2x-thru de-embedding.
- Automated MIPI RFFE compliance validation in PyVisa and SCPI, cutting time from 6 hours to 4 minutes.
- Defined two new validation cases for antenna tuner hubs, improving signal integrity and reducing coupling.
- Automated camera verification over temperature and validated SoC camera clock generation.

MediaTek

Jan 2026 - May 2026

Analog / Mixed-Signal IC Engineer Intern

Woburn, MA

- Led characterization of a zero-IF WiFi-7 transceiver IC: DAC, ADC, front end, and SI performance.
- Optimized RX/TX test benches, saving 2hrs per chip validation and yielding 6dB TX EVM improvement.

Draper

May 2025 - Aug 2025

Systems Engineer Intern

Cambridge, MA

- Characterized an analog ASIC for bench-level power draw and performance across thermal cycling.
- Delivered hardware design and test documentation for a GPS receiver conducting satellite monitoring.

Nokia Bell Labs

Jan 2025 - May 2025

Backscatter Radio Engineer Co-op

Murray Hill, NJ

- Designed 2 PCBs in Altium integrating RF, high speed digital, and low voltage power measurement.
- Automated PCB testbench, reducing bring-up time from 20 minutes to 2 minutes across 300+ devices.
- Cut FPGA power consumption by 15% through System Verilog revisions and synthesis settings.

DEKA Research and Development

May 2024 - Aug 2024

Embedded Software Engineer Intern

Manchester, NH

- Architected C++ modem firmware for latency, fail-over, and security; cut boot time 60% with multithreading.

TECHNICAL PROJECTS

PLL ASIC | Personal Project

May 2025 - Aug 2025

- Designed a 40-100MHz PLL using Sky130 PDK with custom schematic and layout.
- Verified through DRC/LVS checks and parasitic extraction using NGSpice, Xschem, Magic VLSI, Klayout.

16-QAM I/Q Modulator, FPGA | Personal Project

May 2025 - Aug 2025

Implemented a 16-QAM baseband modulator on a Xilinx FPGA in SystemVerilog with a UVM testbench.

LEADERSHIP AND AWARDS

Founder & Director | Forge Silicon Design

Jun 2026 - Current

- Founded university's first chip design student organization, blending silicon with entrepreneurship.

Principal Electrical Engineer | Generate Product Development Studio

Jul 2025 - Current

- Lead 100+ engineers across 8 client projects, owning trade-offs in RF TX, power, & robotic actuators.

Apple Next-gen Innovators Mentorship | 1 of 10 selected

Sep 2025 - May 2026

Silevitch Engineering Leadership Scholarship

Apr 2026

Amazon Edge Technology Scholarship | 1 of 3 selected

Oct 2025